

[« BACK](#)[SHARE](#)[AES \(https://audioxpress.com/tags/aes-audio-engineering-society/\)](https://audioxpress.com/tags/aes-audio-engineering-society/)by [Charlie Hughes \(/authors/23323/charlie-hughes/\)](/authors/23323/charlie-hughes/)on [Theory Articles \(/categories/theory-articles/\)](/categories/theory-articles/) Article[Show more \(12\)](#)

Understanding Horn Directivity Control

🕒 April 7 2015, 07:00



Horns are often used in loudspeaker systems for a variety of reasons. One may be to provide increased acoustical loading to the driver attached to the horn. This helps to increase the sensitivity/efficiency. Another may be to provide some control over the radiation pattern of the driver attached to the horn. It is this directivity control aspect upon which I will focus attention for this article.

In 1975 Don Keele presented a paper at the 51st Convention of the Audio Engineering Society that introduced the modern Constant Directivity horn. This new horn design (new at that time) combined an exponential throat section, to provide good loading at lower frequencies, with a conical or straight wall section that provided a more uniform coverage pattern in the higher frequency region than a pure exponential horn.

In this paper he noted the relationship between the coverage angle of the horn, the low frequency loss of directivity control, and the horn mouth size, so Equation 1 is $d = K/(\Theta \times f\Theta)$.

Where:

d = the mouth dimension of the horn (either horizontal or vertical)

Θ = the coverage angle of the horn (horizontal or vertical)

$f\Theta$ = the low frequency loss of directivity control, Θ (horizontal or vertical) and K is a constant with an empirically derived value or $1 \times 10^6 \times \text{deg} \times \text{Hz}$ of $2.54 \times 10^4 \text{m} \times \text{deg} \times \text{Hz}$.

From this equation you can determine the horn mouth dimension necessary to yield directivity control down to a certain frequency. For a fixed coverage angle, a larger